



inClass- L7

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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L7
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Flow

- Announcement
- Recap
- Module-1 Chapter-2 (M1C2, Random Variables) starts
- 4 lectures planned
 - Discrete rv $\rightarrow$ PMF $\rightarrow$ Bernoulli, Binomial, Geom., Poisson
 - Continuous rv $\rightarrow$ PDF $\rightarrow$ CDF $\rightarrow$ normal distribution
 - Normal, Lognormal, χ^2 , Student-t
 - Expectations $\rightarrow$ variance $\rightarrow$ moments $\rightarrow$ functions of rv

Announcement

- Tutorial-2 today 3—5 pm
 - Venue: **BSB 203**
- Check GitHub regularly

<https://github.com/prakashvs613/2026-CV2090-IITM/>



Messages from the Last Class

- **Message 1: Bayes' Rule**

$$\Pr(\text{guilty} | \text{DNA match}) \propto \Pr(\text{DNA match} | \text{guilty})$$

This is NOT the entire story. 

$$\Pr(G | \text{DNA match}) \propto \Pr(\text{DNA match} | G) \cdot \Pr(G)$$

In other words,

probability $\propto$ Likelihood $\leftarrow$ *Misleading*

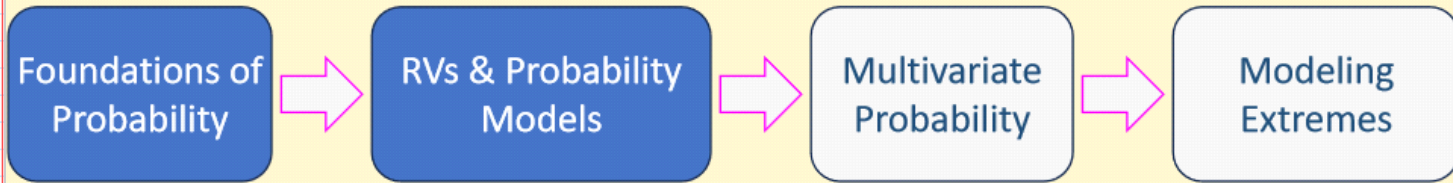
~~probability~~ $\propto$ Likelihood $\times$ prior $\leftarrow$ *Appropriate*

Posterior $\propto$ Likelihood $\times$ prior

Posterior does not exist in isolation

Course Flow

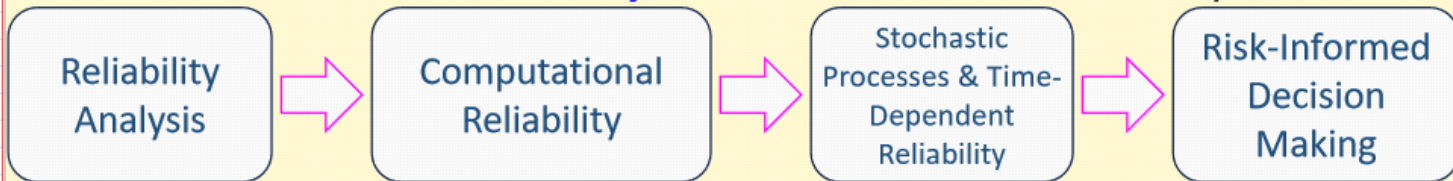
Module 1: **Probability** – Understanding Uncertainty



Module 2: **Statistics** – Learning from Data

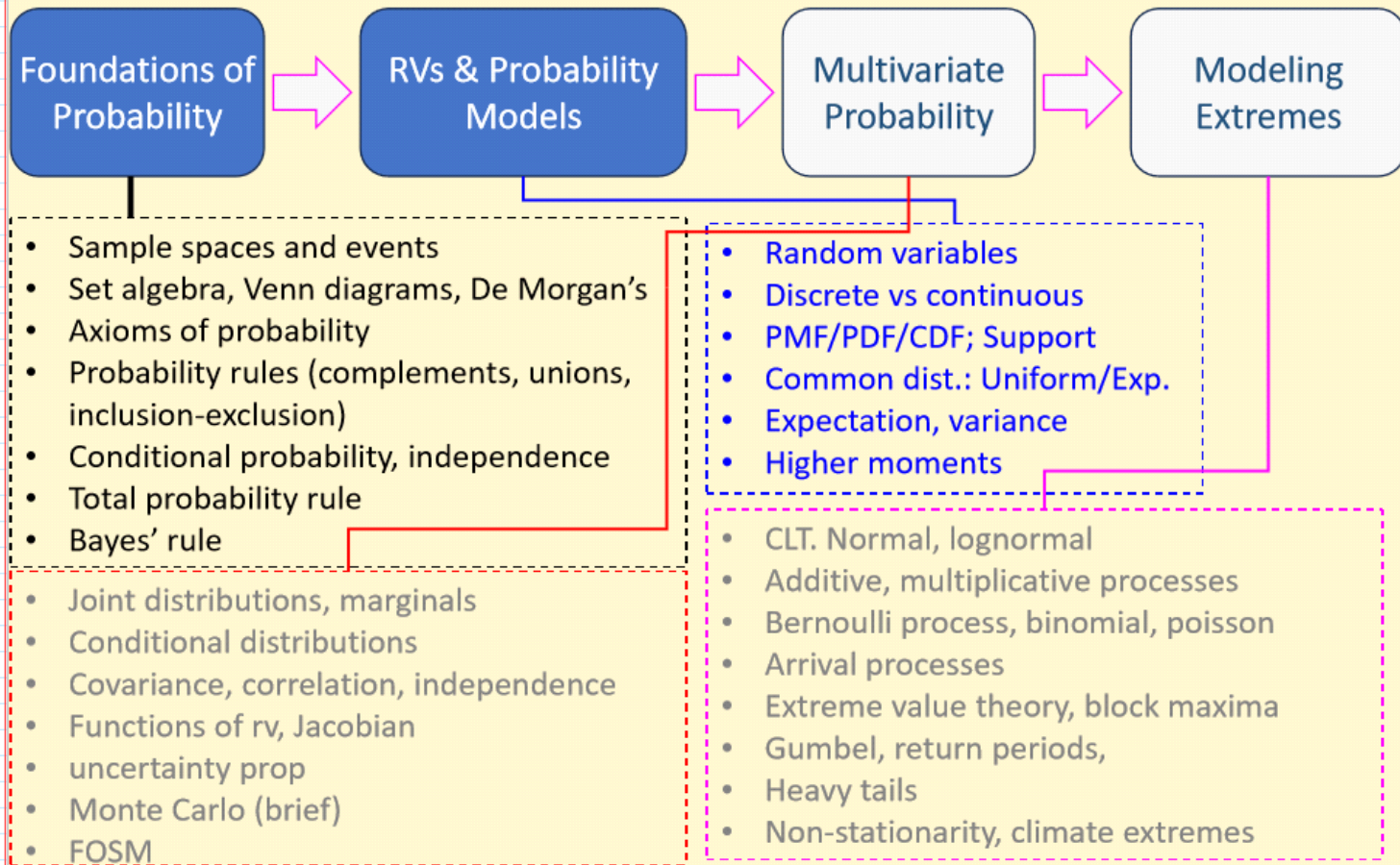


Module 3: **Reliability** – Decision under Uncertainty



Module I: Chapter 2 (M1C2)

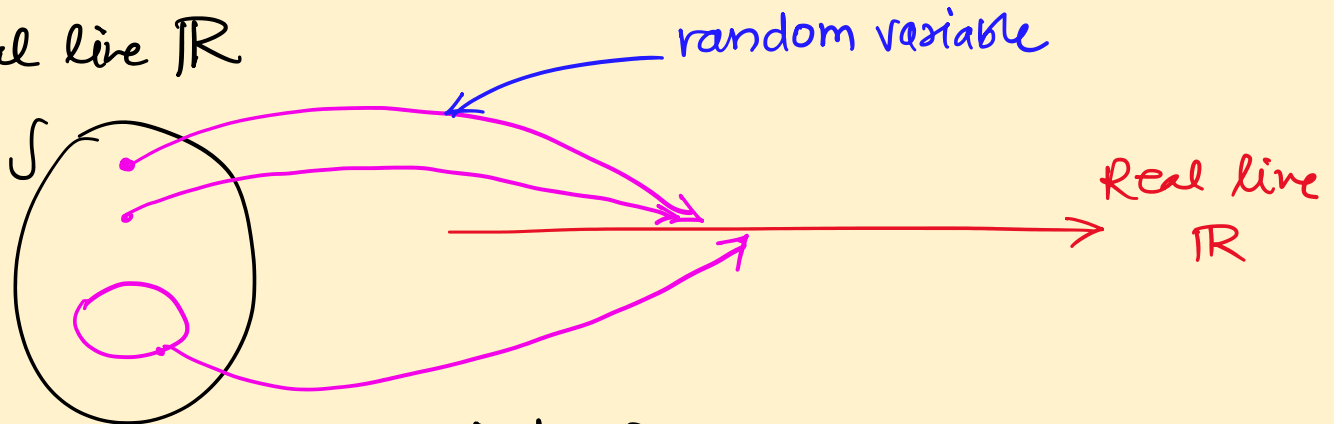
Module 1: **Probability** – Understanding Uncertainty



Random Variables

Random variable

→ Given a (random) experiment and a sample space S , a random variable (rv, RV) is a function from S to real line $\mathbb{R}$



→ Experiment: Two coin tosses
 $S = \{HH, HT, TH, TT\}$ $X = \text{number of heads}$

$$X(\omega) = \begin{cases} 0, & \text{if } \omega = \{TT\} \\ 1, & \text{if } \omega = \{TH\}, \{HT\} \\ 2, & \text{if } \omega = \{HH\} \end{cases}$$

$\omega \in S$
 $X(\omega) \in \mathbb{R}$

$\uparrow$
 a random variable

A misnomer;
 neither random
 nor a variable.

Notation:
 UPPERCASE
 LETTER, RV
 lowercase,
 outcome of a RV.

Random variable

- Experiment: rolling two 4-faced dice

X = maximum roll

Domain of X ?

Range of X ?

$S = ?$

$X(w) = ?$

Remarks

- ① A real-valued function of an experiment $\rightarrow$ RV
- ② A function of RVs is also a random variable.
- ③ A RV can be conditioned on an event.
- ④ A RV can also be conditioned on another RV.

Types of Random Variables

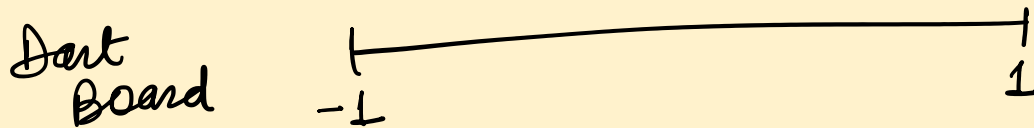
Types of RV

→ Discrete

- Range is finite or countable
e.g; your grade in CV2090,
proportion of dented chips in a lot of 1000,

→ Continuous

- Range is continuous over some interval
e.g; height of all Indians,



Experiment: Dart throw on a $[-1, 1]$ board.

PMF

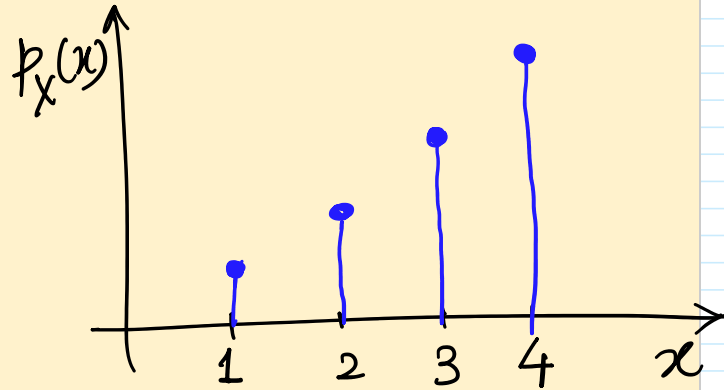
Probability Mass Function (PMF)

→ For a discrete RV, a probability mass function (PMF) is defined as

PMF → $p_X(x) = \Pr(X=x)$

Write down the PMF of $X =$ maximum roll in a two 4-faced dice roll

$$p_X(x) = \begin{cases} 1/16, & x=1 \\ 3/16, & x=2 \\ 5/16, & x=3 \\ 7/16, & x=4 \end{cases}$$



Probability Mass Function (PMF)

Properties

1. $p_X(x) = \Pr(X=x)$

2. $0 \leq p_X(x) \leq 1$

3. $\sum_x p_X(x) = 1$

Class problem on PMF

X is the maximum value from rolling two 4-faced dice, determine $p_X(x)$.

Some Common Discrete Distributions

Bernoulli rv

→ Takes only two values
0, 1

e.g. coin toss: Head as 1
fail/pass:

→ Single parameter RV: p

→ PMF of a Bernoulli rv is

$$p_X(x) = \begin{cases} p, & x=1 \\ 1-p, & x=0 \end{cases}$$

for a fair coin, $p = 1/2$

for a biased/loaded coin, $p \neq 1/2$

Binomial rv